Gavin Reis

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Education

Cornell University, College of Engineering Ithaca, NY

B.S. Electrical and Computer Engineering, Minor in Artificial Intelligence *Expected May 2028*

Coursework: Embedded Systems (ECE 3140), Digital Logic & Computer Organization (ECE 2300), Signals & Systems (ECE 3250), Foundations of AI Reasoning (CS 3700), Data Science for Engineers (ECE 2720), OOP & Data Structures (ENGRD 2110), EM Fields & Waves (ECE 3030), Linear Algebra (MATH 2940)

Experience

Embedded Intent Long Island, NY

Founder & Engineer, web development and AI automation agency *2026 – Present*

- Run a one-person agency that builds websites and lead-capture automations for local service businesses: finding clients, scoping, pricing, design, build and delivery; wrote and priced 11 productized services, with monthly care, SEO and automation plans.
- Built and shipped embeddedintent.com (Next.js 16, TypeScript strict, Tailwind v4, Vercel) with a one-command quality gate every pull request must pass: typecheck, lint, production build, horizontal-overflow checks at 360/390/430 px, full-page screenshots and Lighthouse mobile on all 22 routes (latest run: 100 accessibility and best practices, zero layout shift on every route).
- Typed case studies as a discriminated union so the build fails if a result is shown without a measured before, after and source.
- Building a realtor client's site under an $8,000 engagement, with a live listings pipeline (PostgreSQL via Drizzle on Neon, image processing with sharp, Vercel Blob storage) and per-area landing pages; launch pending.

Above All Tents Long Island, NY

Web Developer & Operations Lead *May 2026 – Aug 2026*

- Rebuilt abovealltents.com from the ground up under a $3,000 engagement (Next.js, TypeScript, Tailwind CSS, Vercel), mobile-first, around a quote request that captures event date, guest count and occasion up front, with call and quote buttons pinned on phones; launched August 2026.
- Added LocalBusiness structured data (JSON-LD with hours, location and price range) and semantic markup for local search.
- Scheduled and dispatched a 10-person install crew across roughly 12 events per week; primary client contact for quoting, scheduling and on-site walkthroughs.

Projects

HQ: Multi-Agent Operations System | *Node.js, SQLite, tmux, Claude Code, OpenAI Codex* 2026

- Built the system that runs my agency's AI staff: a team of agents (a coordinator, four managers, a secretary, builders and a Codex advisor) running as persistent sessions, with work levels that cap agents, models and concurrent work.
- Designed a durable work-order store on SQLite with claims, leases, attempt-bound reports (a timed-out worker can't complete a retried task) and approvals bound to one request and its action hash, with an append-only event log.
- Wrote a deterministic scheduler and watchdog (no model calls) with time-zone-aware catch-up, quiet hours and deduplicated phone alerts, plus a local dashboard for answering agent prompts and reading their transcripts.
- Covered by 120+ automated tests using stand-in agents, isolated tmux servers and headless Chromium; every change ships through a pull request with an independent AI code review.

32-Bit RISC-V Processor & Digital Design Suite | *Verilog, FPGA, RISC-V Assembly* ECE 2300, Fall 2025

- Designed and verified a single-cycle 32-bit TinyRV1 processor in Verilog (ADD, ADDI, MUL, LW, SW, JAL, JR, BNE) with a custom control unit, 32×32b register file, ALU, immediate generator and multiplier.
- Built the arithmetic units from the gate level (16-bit carry-select adder, 2×16b multiplier, load/enable counter) and compared ripple-carry against carry-select on critical path, area and delay.
- Wrote RISC-V assembly on the FPGA over memory-mapped I/O (a button-driven calculator and a door-activity monitor with a timed buzzer), with exhaustive and X-propagation testbenches for every module.

Autonomous PacMann Agent | *Java, JUnit, Swing* ENGRD 2110, Spring 2026

- Built a 2,900-line MVC game over a weighted, wraparound maze graph with Dijkstra's shortest path under a non-backtracking constraint, and a heuristic agent scoring moves on pellet proximity, ghost distance and chase/flee state.
- Implemented four ghost pursuit strategies (direct chase, lookahead, vector reflection, distance-gated retreat), validated with 18 JUnit tests.

Analog Signal-Chain Receiver | *LTspice, oscilloscope, function generator* ECE 2100, Fall 2025

- Designed a 4th-order multiple-feedback bandpass filter with a trimmer-tuned center frequency and an envelope detector that recovers an encoded audio message; verified the measured response against LTspice.

Technical Skills

Languages: Python, Java, TypeScript, JavaScript, C, Verilog, ARM Assembly, SQL, HTML/CSS, MATLAB
Frameworks: Next.js, React, Node.js, Tailwind CSS, Drizzle ORM, PyTorch, NumPy, pandas, scikit-learn, JUnit
AI & tooling: Claude Code, OpenAI Codex, multi-agent orchestration, Git/GitHub, Linux, tmux, Vercel, Lighthouse, Quartus, LTspice
Hardware: NXP FRDM-KL46Z (ARM Cortex-M0+), FPGA boards, oscilloscopes, logic analyzers, function generators, breadboarding